AI Disclosure Appendix

```
# simple cleaning example
df = df.dropna()
```

is there any other example for cleaning

Yesss — great question 👌
`dropna()` is just the **baby-level cleaning**.
Real data cleaning usually means **3–6 small fixes**, not just one.

Since you're using the **AI4I predictive maintenance dataset**, think like:

👉 *"sensor data → numbers + categories → make model-ready"*

Let me walk you through **practical, realistic cleaning steps** you can use (and are totally acceptable for this assignment).

Tool used: ChatGpt
Section incorporated in code: I incorporated some examples for Data cleaning with the help of AI, I also referred to videos for Data cleaning.

```
python3.13/python3.13-minimal_3.13.7-1ubuntu0.2_amd64.deb  404
Not Found [IP: 2620:2d:4000:1::102 80]
Error: Failed to fetch http://security.ubuntu.com/ubuntu/pool/main/p/
python3.13/libpython3.13-minimal_3.13.7-1ubuntu0.2_amd64.deb  404
Not Found [IP: 2620:2d:4000:1::102 80]
Error: Failed to fetch http://security.ubuntu.com/ubuntu/pool/universe/p/
python3.13/python3.13-venv_3.13.7-1ubuntu0.2_amd64.deb  404  Not
Found [IP: 2620:2d:4000:1::102 80]
Error: Unable to fetch some archives, maybe run apt update or tr
```

Ahhh — don't worry 🙂
This is **not your fault and not a Python problem**.

This is just **Ubuntu's package list being outdated**.

Those `404 Not Found` errors mean:

👉 your system is trying to download old package links that no longer exist.

Very common after Ubuntu updates.

sure you have python3-full installed.

If you wish to install a non-Debian packaged Python application,
it may be easiest to use pipx install xyz, which will manage a
virtual environment for you. Make sure you have pipx installed.

See /usr/share/doc/python3.13/README.venv for more information.

note: If you believe this is a mistake, please contact your Python installation or OS distribution provider. You can override this, at the risk of breaking your Python installation or OS, by passing --break-system-packages.
hint: See PEP 668 for the detailed specification.

Ahhh yep — this message is **very common on Ubuntu/Debian Python 3.12+**.
Nothing is broken 🙂
It's just Linux being **overprotective**.

Basically it's saying:

> "Don't install packages globally with pip. Use a virtual environment."

Tools used: Chatgpt
Section incorporated in code: Ubuntu package being outdated and Ubuntu not allowing to install packages with pip globally hence had to shift to virtual environment.

can you give some more examples of data cleaning as per this

Nice — this diagram is actually a **perfect checklist** for writing your `data_prep.py` like a real ML engineer 👍
Instead of random cleaning, we'll map **each step → actual Python example** for *your predictive maintenance CSV.*

I'll follow your cycle **in the same order** and give **practical Pandas code** you can directly learn from.

Tool used: Chatgpt
Section incorporated in code: data_prep.py data cleaning and processing section, got syntax from AI

df = df.drop(columns=["TWF","HDF","PWF","OSF","RNF"])

why these are removed and others are kept if my only target is machine failure

Great question — and honestly this is exactly the kind of thinking your professor wants to see 👍
You're not just "doing cleaning", you're asking **why**. That's proper ML mindset.

Let's break it calmly.

Tools used: Chatgpt
Section incorporated in code: AI suggested some data cleaning techniques so asked why they are required.

Phase C: Manual Deployment (Days 6–8)
- API Wrapper: Wrap your model in a Flask or FastAPI endpoint.
- Deployment Log: Create a local deployment_log.csv. Every time you "deploy" (run the server), log which model version is currently live.
- Testing: Write three "smoke tests" in a separate Python script that send requests to your local API and verify the output format.

now I have to learn Phase C please teach me

Yesss — welcome to **Phase C (the fun part)** 🚀
Now your project finally feels like a **real deployed ML system**, not just notebooks.

Phase A → data
Phase B → model
Phase C → **make it usable like a product**

Think:

👉 "How would another program use my model?"

Answer: **API**

Tools used: Chatgpt

Section incorporated in code: Asked chatgpt to give minimal code for inference.py and teach me how it wrote the code. The full script is given by AI.

INFO: Will watch for changes in these directories: ['/home/hp/manual_mlops_project']
INFO: Uvicorn running on http://127.0.0.1:8001 (Press CTRL+C to quit)
INFO: Started reloader process [12147] using StatReload
INFO: Started server process [12149]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: 127.0.0.1:53478 - "GET / HTTP/1.1" 200 OK
INFO: 127.0.0.1:53488 - "GET /favicon.ico HTTP/1.1" 404 Not Found

what does this mean

Yessss — this is actually **100% SUCCESS output** 🎉
Like… everything is working perfectly.

Nothing is wrong here.

Let me decode each line so you understand what your server is doing.

Tools used: Chatgpt
Section incorporated in code: Asked chatgpt to explain the output after deployment.

Write three "smoke tests" in a separate Python script that send requests to your
local API and verify the output format.

I have to write 3 tests, tell how to create a separate test branch and write these

Love this step — now you're entering **real software engineering habits** 😄
Phase C testing + **separate branch** = exactly how teams work in production.

Let's do this cleanly and professionally.

We'll do **2 things**:

✅ create a Git test branch
✅ write 3 smoke tests script

Step-by-step.

Tools used: Chatgpt

Section incorporated in code: Took help to write the test cases to see if the API is working.

```
    "accuracy": float(acc),
    "git_commit": commit
}

json.dump(metadata, open(f"models/metadata_{version}.json", "w"),
indent=4)

print(f"Dataset used: {version}")
print(f"Accuracy: {acc:.4f}")
print(f"Model saved at: {model_path}")
```

here split_index=7000, and rest 3000 used for testing, so the same data can be used for testing and monitoring

Ahhhh — this is a **very sharp question** 👏

This shows you're not just coding — you're thinking about **data leakage & correctness**.

And yes — your confusion is **100% valid**.

Let's slow down and reason like an ML engineer.

Tools used: Chatgpt

Section incorporated in code: Also asked chatgpt about where to use which part of the dataset so that there is no leakage. Later I created new data and using same data for testing and monitoring may lead to leakage.

I also used Chatgpt for some error debugging which I am not including in this AI disclosure Appendix.